Milestone 1 Report
EDA and Hypothesis Testing

DSA 210 Project

1 Project Idea

In this project, I study which factors may be related to university students' academic performance. The main goal of this milestone is not full prediction yet. The goal here is to understand the data first, make a basic exploratory analysis, and test a few early statistical hypotheses before moving to machine learning in the next milestone.

2 Dataset and Data Collection

I used the *Higher Education Students Performance Evaluation* dataset. The file used in this milestone is `student_performance_value.csv`. The dataset contains 145 student records and 32 columns. The target variable is `output_grade`. This variable represents the final course outcome as categorical grade labels such as Fail, DD, DC, CC, CB, BB, BA, and AA.

The dataset also contains many explanatory variables such as sex, scholarship type, study hours, attendance to classes, note-taking behavior, listening in class, and cumulative GPA from the previous semester. This makes the dataset suitable for a multivariable project.

3 EDA Process

In the EDA part, I first checked the shape of the dataset, the column names, the data types, and the missing values. After that, I looked at the category counts of important variables such as output grade, study hours, attendance to classes, previous semester GPA, sex, and scholarship type.

Then I created simple visualizations to understand the data more clearly. First, I plotted the distribution of the output grades. This gave a quick view of the class balance in the target variable. Then I created stacked bar charts for weekly study hours versus output grade, attendance to classes versus output grade, and previous semester GPA versus output grade.

The grade distribution shows that DD is the most common category in this dataset, while Fail is the least common category. The study-hours plot suggests that most students are in the `<5 hours` group. The attendance plot shows that most students report `always` attending classes. The previous GPA plot suggests that students with stronger GPA groups tend to appear more often in the higher output-grade categories.

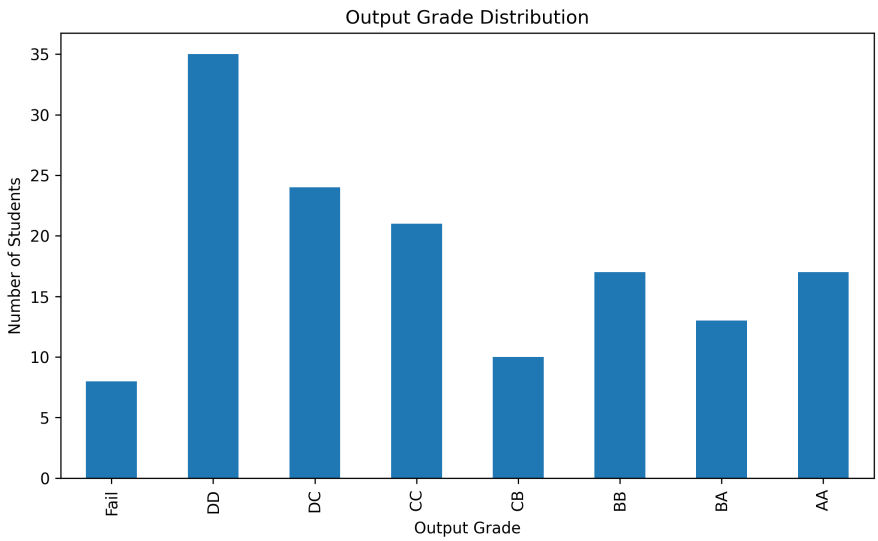

Figure 1: Distribution of the output grade variable.

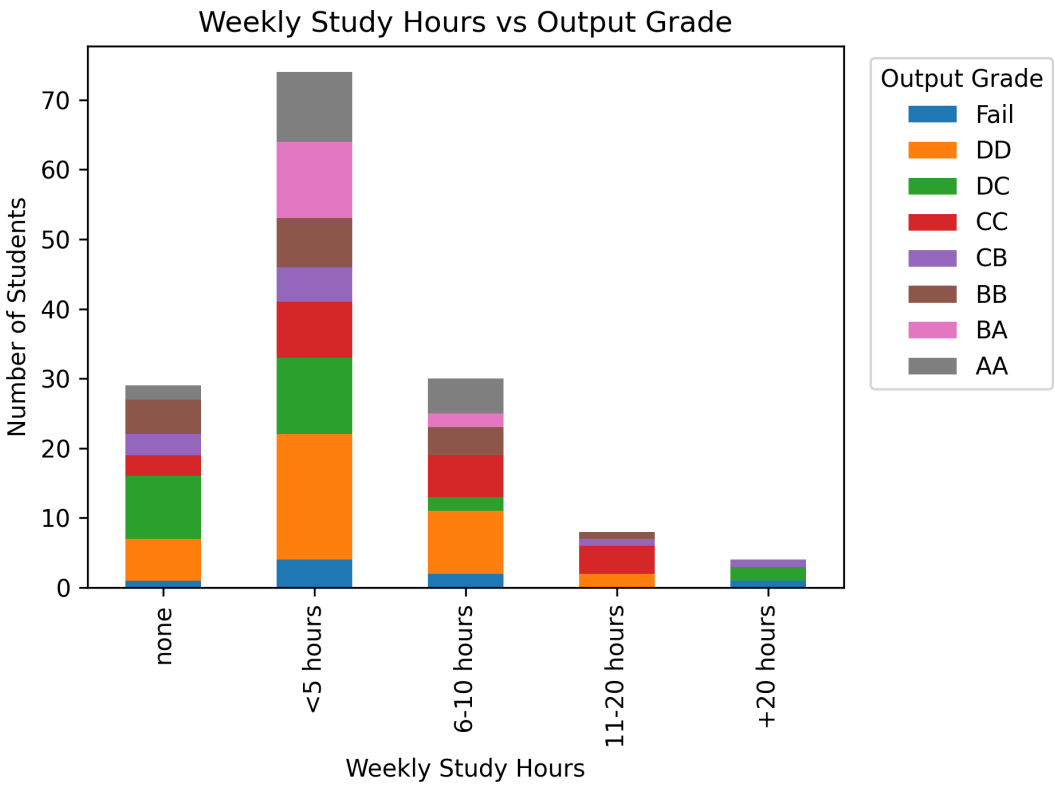

Figure 2: Weekly study hours and output grade.

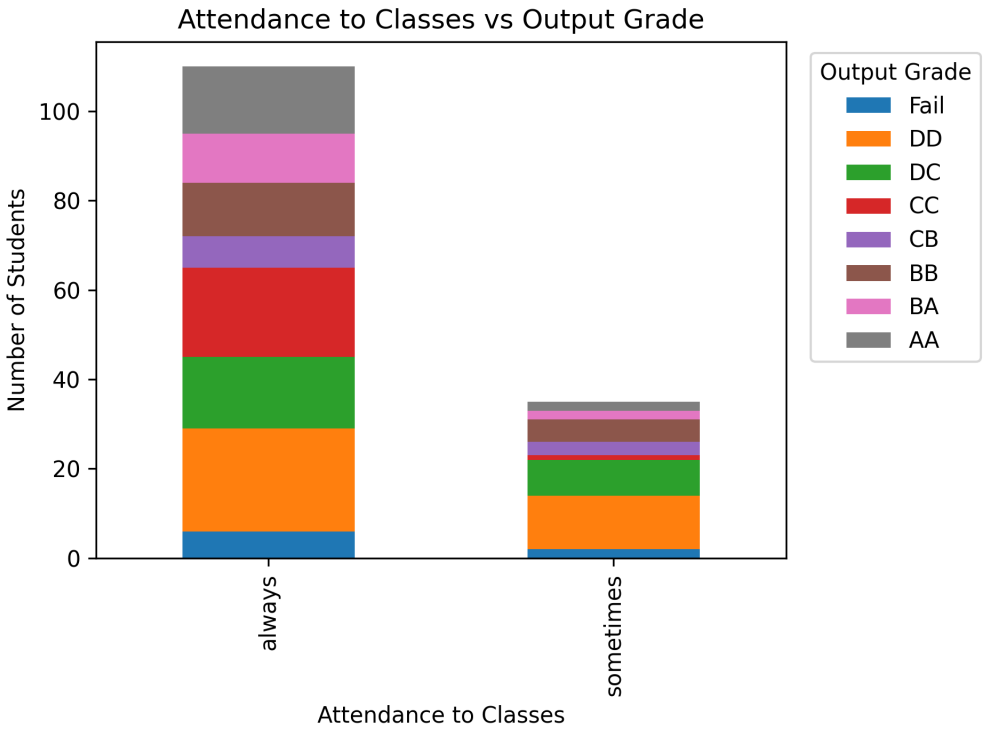

Figure 3: Attendance to classes and output grade.

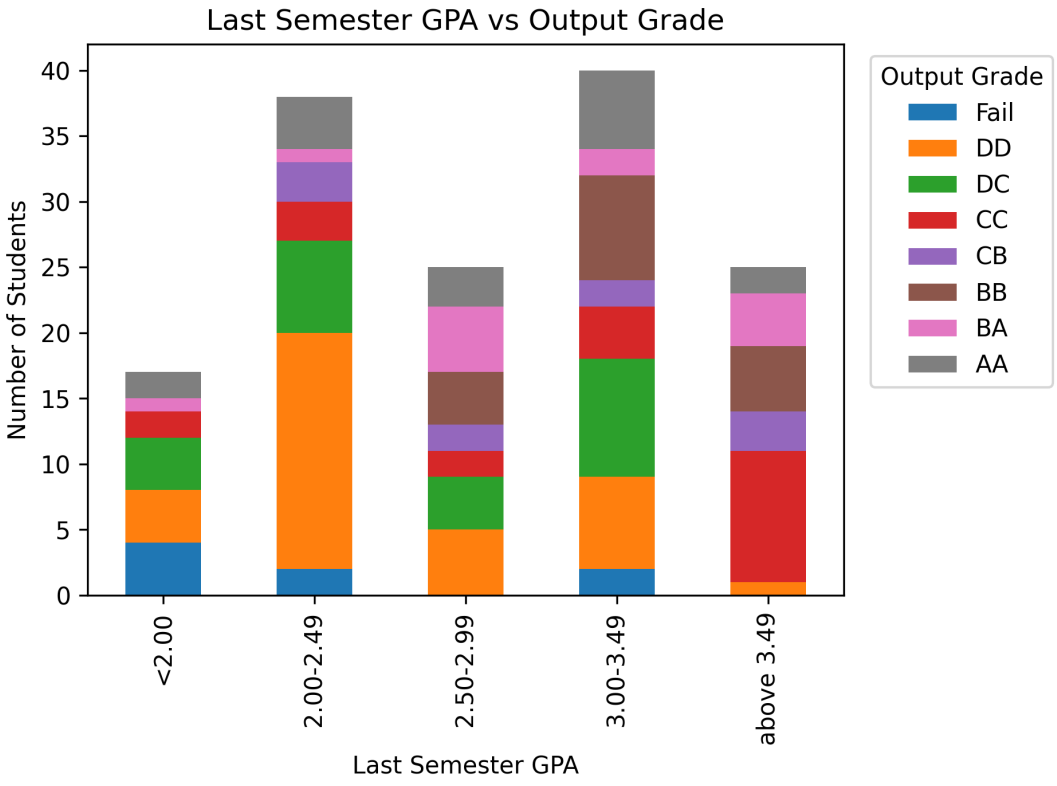

Figure 4: Last semester GPA and output grade.

4 Hypotheses and Tests

I used several hypothesis tests in this milestone. Because many variables are categorical or ordinal, I selected tests that fit this structure.

4.1 Hypothesis 1: Attendance and Output Grade

H0: `attendance_to_classes` and `output_grade` are independent.
H1: `attendance_to_classes` and `output_grade` are related.

I used a chi-square independence test. The p-value was 0.200408. Since this value is greater than 0.05, I fail to reject the null hypothesis. In this sample, there is not enough evidence to claim a statistically significant relationship between attendance category and output grade.

4.2 Hypothesis 2: Weekly Study Hours and Output Grade

H0: `weekly_study_hours` and `output_grade` are independent.
H1: `weekly_study_hours` and `output_grade` are related.

I again used a chi-square independence test. The p-value was 0.068471. This result is close to 0.05 but still above the threshold. Because of that, I fail to reject the null hypothesis at the 5% significance level. This may still be worth checking again later with different modeling choices.

4.3 Hypothesis 3: Previous Semester GPA and Output Grade

H0: Last semester GPA and output grade are not associated.
H1: Last semester GPA and output grade are associated.

For this part, I converted the ordered GPA groups and grade labels into ordinal scores and used Spearman correlation. The correlation was 0.3508 and the p-value was 0.000015. This is clearly below 0.05, so I reject the null hypothesis. This is the strongest result in the milestone. It suggests that students with higher GPA groups from the last semester also tend to have better output grades in the current course.

4.4 Hypothesis 4: Grade-Score Differences Across Study-Hour Groups

H0: The grade-score distributions are the same across study-hour groups.
H1: At least one study-hour group has a different grade-score distribution.

I used the Kruskal-Wallis test because the grade scale is ordinal after mapping. The p-value was 0.759686. This is much larger than 0.05, so I fail to reject the null hypothesis.

5 Main Findings

The strongest finding from this milestone is that previous semester GPA has a meaningful positive relationship with output grade. This is both visually reasonable in the EDA and statistically supported by the hypothesis test. Attendance and study-hour variables are interesting, but in the current tests they do not reach the standard significance threshold.

6 Limitations

This dataset is small, with only 145 observations. Also, many variables are categorical and self-reported, which may limit precision. Another limitation is that the current milestone uses simple univariate and bivariate analysis. At this stage, I am not yet controlling for several variables at the same time.

7 Next Step

In the next milestone, I plan to convert the categorical variables into model-ready features and apply machine learning methods for prediction of output grade. I will compare multiple models and evaluate which features are most useful for prediction.